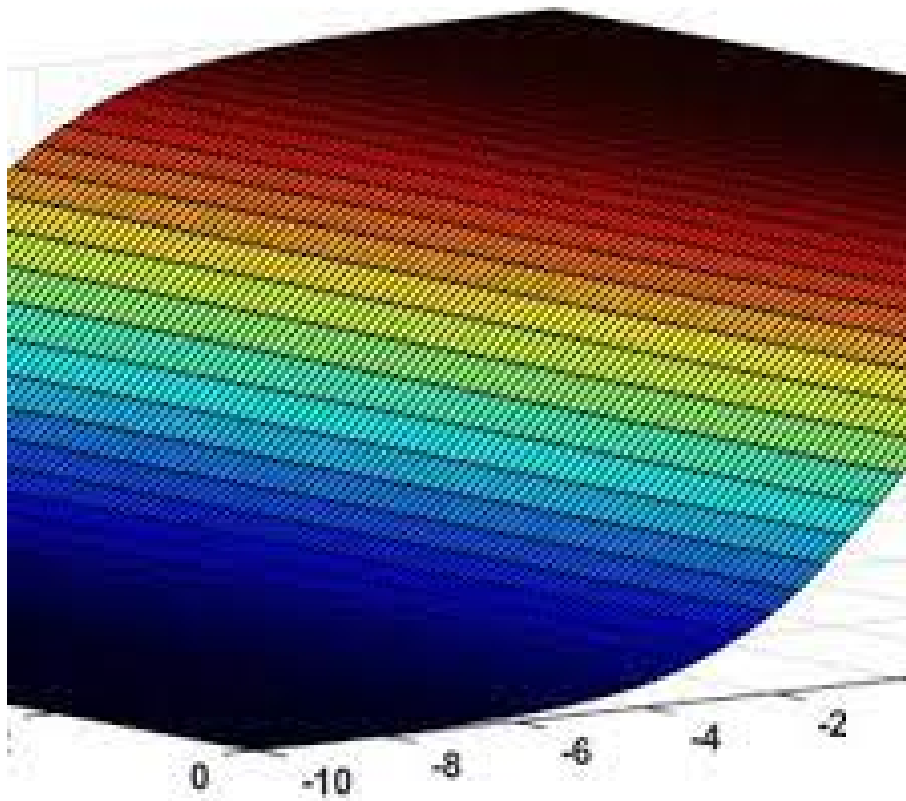


FHN Model Report

Shivan Ajay Iyer



Overview

This report details the simulation of the FitzHugh-Nagumo (FHN) neuron model. The goal was to explore the model's behaviour under different levels of external stimulus current (I_{ext}) by solving its governing differential equations numerically. The simulations were performed using the forward Euler method to integrate the equations over time, and the system's dynamics were analysed using phase plane plots and time-series graphs

Logic and Methodology

Model Equations

The FHN model is defined by two coupled first-order differential equations for the membrane voltage (v) and a recovery variable (w).

$$\text{Voltage Equation: } \frac{dv}{dt} = v(a - v)(v - 1) - w + I_{ext}$$

$$\text{Recovery Equation: } \frac{dw}{dt} = bv - rw$$

The base parameters used were $a = 0.5$, $b = 0.1$, and $r = 0.1$, except in Case 4, where b and r were modified to demonstrate bistability.

Phase Plane Analysis

The core logic for analysing the system's behaviour involves the nullclines

1. V-Nullcline ($dv/dt = 0$):

$$w = v(a - v)(v - 1) + I_{ext}.$$

2. W-Nullcline ($dw/dt = 0$)

$$w = \frac{b}{r}v$$

The **fixed points** of the system are the intersections of these two nullclines. The stability of these points, along with the vector field (visualised in the phase plots), determines the system's long-term behaviour (e.g., whether it settles at a fixed point or enters a limit cycle).

The stability of the fixed points can be determined by the eigenvalues of the Jacobian that describes the system at the respective points.

$$J(v, w) = \begin{pmatrix} -3v^2 + 2(a+1)v - a & -1 \\ b & -r \end{pmatrix}$$

Key Observations from Simulation Cases

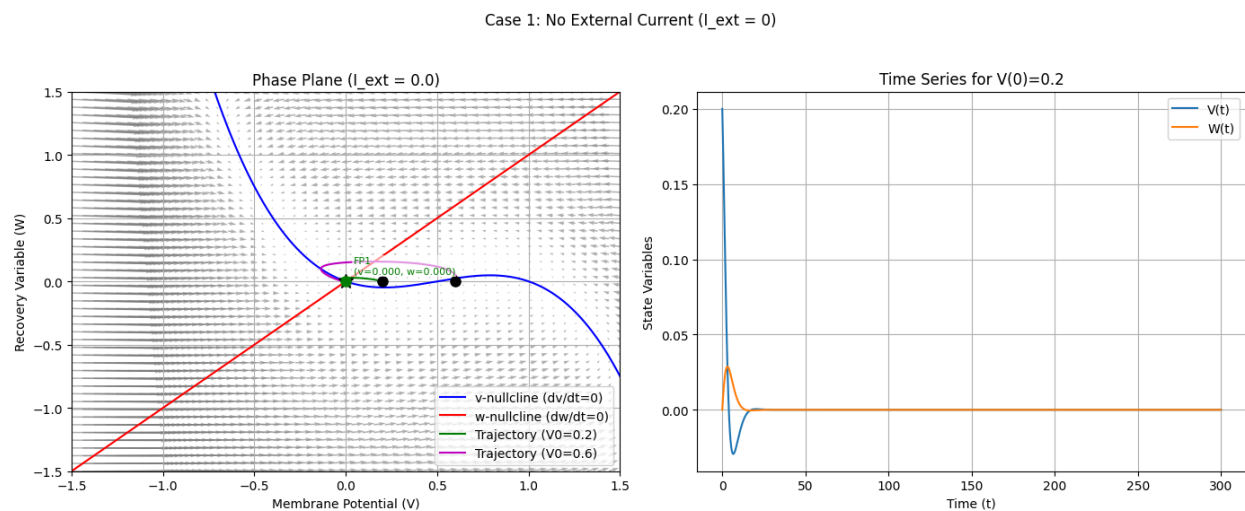
Case 1: No External Current ($I_{\text{ext}} = 0$)

This case simulates the neuron's "resting" state with no external input. Trajectories were simulated for initial conditions $V(0) < a$ and $V(0) > a$.

Observations:

- The nullclines intersect at a single fixed point near the origin (0,0), which is stable
- Both trajectories, regardless of whether $V(0)$ was 0.2 or 0.7, were attracted to this single stable point.
- The time-series plot shows that for $V(0) = 0.7$, the voltage makes a single "excursion" (a spike) before damping down to the resting state. For $V(0) = 0.2$, it returns directly to rest. This demonstrates the model's excitability—it can fire a single "action

potential" in response to a super-threshold stimulus but returns to rest without repetitive firing



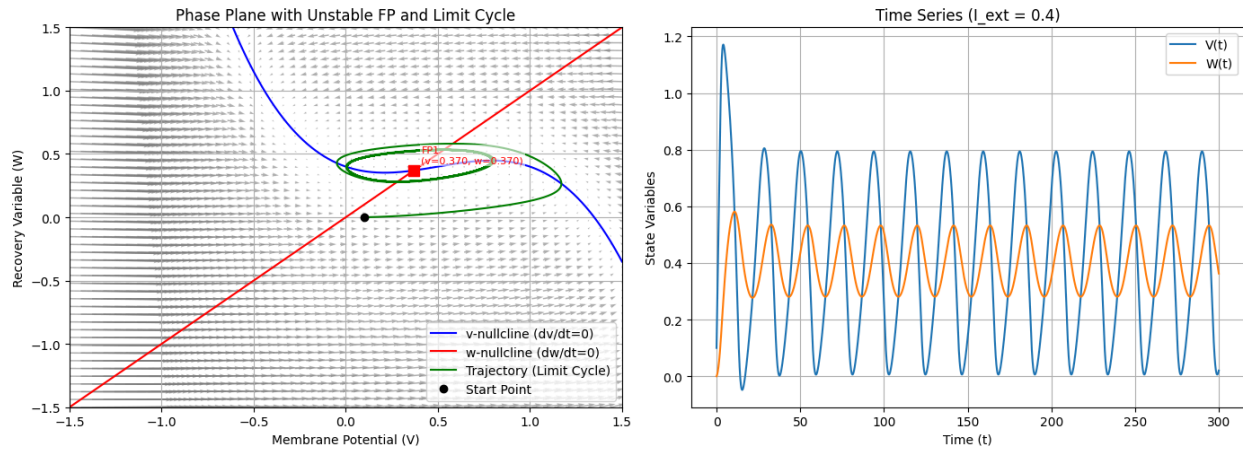
Case 2: Oscillations ($I_{\text{ext}} = 0.4$)

This case demonstrates the neuron's repetitive firing state, achieved by applying an intermediate constant current.

Observations:

- With $I_{\text{ext}} = 0.4$, the single fixed point (intersection) moves up and to the right along the cubic nullcline.
- This fixed point is now **unstable**.
- The phase plot clearly shows that the trajectory, starting from $(0.1, 0)$, does not settle at the fixed point. Instead, it spirals outwards and is attracted to a stable limit cycle
- This limit cycle represents the repetitive, periodic firing of action potentials, which would be seen as regular oscillations on a $V(t)$ time-series plot.

Case 2: Oscillatory Regime



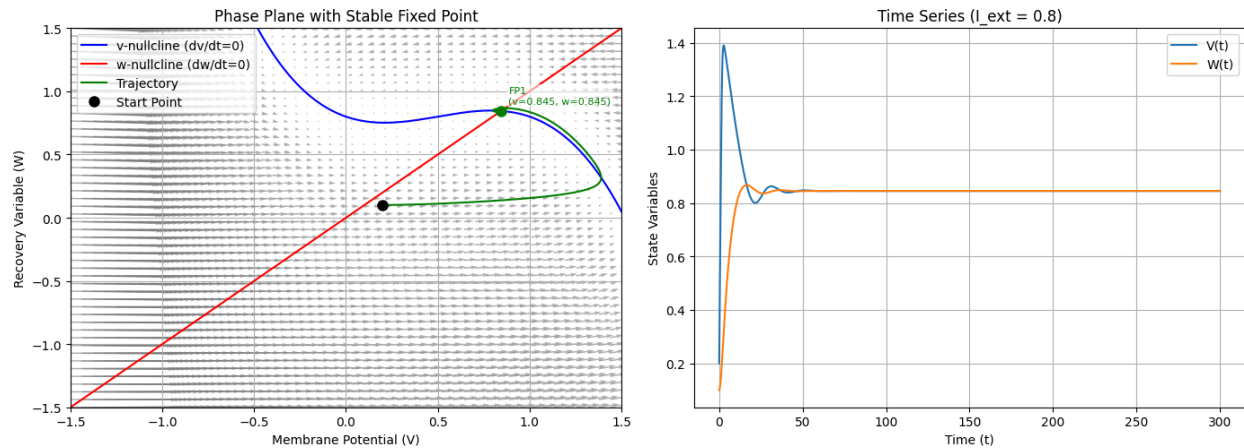
Case 3: High External Current ($I_{\text{ext}} = 8.0$)

This case simulates the effect of a strong, sustained input current

Observations

- For $I_{\text{ext}} = 0.8$, the fixed point is pushed even further to the right, landing on the far-right branch of the cubic nullcline.
- This fixed point is stable.
- The phase plot shows the trajectory being directly attracted to this high-voltage stable point.
- This represents a state of **depolarisation block**, where the external current is so strong that it "locks" the neuron in a high-voltage state, preventing it from repolarising and firing any further action potentials.

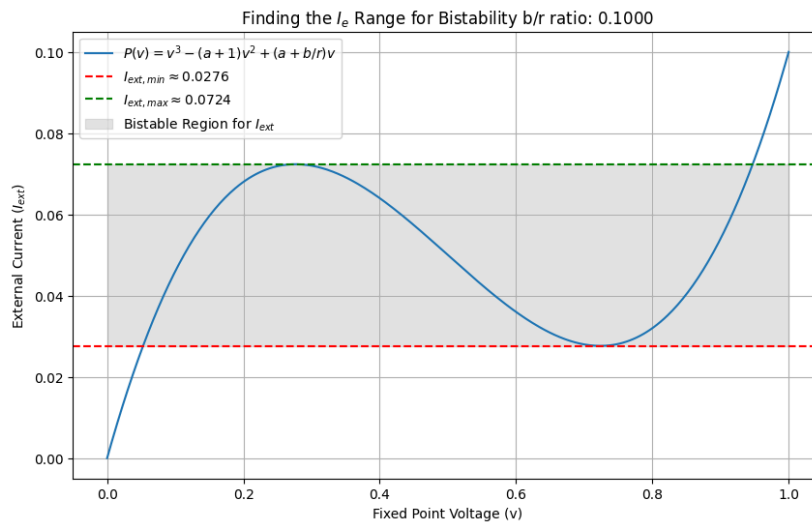
Case 3: High Current, Stable State



Case 4: Bi-stability ($I_{ext} = 0.05$, $b = 0.01$, $r = 0.1$)

Bistability refers to the phenomenon wherein a system has two stable states separated by an unstable state. This property allows a neuron to use a brief stimulus to switch between states.

This case required modifying the model parameters to $b/r < 0.25$ (here, $0.01 / 0.1 = 0.1$) to make bi-stability possible.



A specific $I_{ext} = 0.06$ was then chosen to place the system within its bi-stable range.

Observations

- The nullclines now intersect at three fixed points.
- The phase plot and trajectories confirm the stability of these points: the leftmost (P1) and rightmost (P3) points are stable, while the middle point (P2) is unstable (a saddle point).
- The system now has two distinct stable states. The trajectory labelled "Trajectory to P1" settles at the lower resting state, while the "Trajectory to P3" settles at the higher, more excited state.
- The "Trajectory from P2" shows that any small perturbation from this unstable point will cause the system to be pushed away and fall into one of the two stable "basins of attraction" (in this case, P3).
- This demonstrates that the system's final state is dependent on its initial conditions, the hallmark of bistability.

Case 4: Bistability ($I_{\text{ext}}=0.06$, $b/r=0.10$)

